

LIGHTWEIGHT ZERO-DAY ATTACK DETECTION FOR IOT EDGE DEVICES

**A Project Work Submitted in Partial Fulfillment of
the Requirements for the Degree of Bachelor of
Technology in Computer Science & Engineering**

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CERTIFICATE OF APPROVAL

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ABSTRACT

The rapid growth of Internet of Things (IoT) devices has introduced significant cybersecurity challenges in modern network environments. Traditional intrusion detection systems mainly depend on signature-based techniques, which are effective only for previously known attacks and often fail to identify unseen or zero-day attacks. Moreover, many advanced deep learning-based security solutions require high computational resources, making them unsuitable for lightweight IoT edge devices.

This project proposes a lightweight hybrid intrusion detection framework for IoT edge environments using Machine Learning and Deep Learning techniques. The proposed system combines LightGBM and Autoencoder models to achieve efficient detection of both known and unknown cyberattacks. LightGBM is used for supervised classification of known attack classes, while the Autoencoder model performs anomaly detection by learning normal traffic behavior and identifying abnormal patterns as potential zero-day attacks.

The Edge-IIoT dataset is used for experimentation. Data preprocessing includes cleaning, duplicate removal, feature normalization, encoding, and separation of known and unseen attack classes. Experimental results show that the LightGBM model achieves very high accuracy for known attack classification, while the Autoencoder effectively detects anomalous network traffic. A fusion mechanism combines both approaches to create a lightweight and balanced intrusion detection system suitable for IoT edge deployment.

The final system demonstrates strong performance in detecting both known and zero-day attacks while maintaining low computational complexity and efficient processing capability.

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1. INTRODUCTION

1.1 Background

The Internet of Things (IoT) refers to a network of interconnected devices capable of collecting and exchanging data through the internet. IoT technology is widely used in smart homes, healthcare systems, industrial automation, transportation systems, agriculture, and surveillance systems. Due to the increasing deployment of IoT devices, network traffic has grown rapidly, leading to major cybersecurity concerns.

IoT edge devices usually operate with limited computational power, storage, and memory resources. Traditional security mechanisms designed for powerful servers are often unsuitable for edge environments because they consume high computational resources. Furthermore, cyberattacks targeting IoT networks are becoming increasingly sophisticated.

Common attacks in IoT environments include:

- Distributed Denial of Service (DDoS)
- Man-in-the-Middle (MITM)
- SQL Injection
- Password attacks
- Port scanning
- Ransomware
- Botnet attacks
- Backdoor attacks

Traditional intrusion detection systems primarily rely on predefined signatures and attack patterns. These systems perform effectively for known attacks but struggle to identify new or unseen attack behaviors known as zero-day attacks.

Machine Learning and Deep Learning techniques provide intelligent approaches for identifying malicious network behavior automatically. However, many existing deep learning approaches require high computational complexity and are unsuitable for lightweight IoT edge deployment.

This project focuses on developing a lightweight hybrid intrusion detection system using LightGBM and Autoencoder models for efficient known and zero-day attack detection in IoT edge environments.

1.2 Problem Statement

Most traditional intrusion detection systems fail to detect unknown or zero-day attacks because they depend heavily on previously known attack signatures. Existing deep learning-based cybersecurity systems often require high computational resources and are difficult to deploy on lightweight IoT edge devices.

Therefore, there is a need for a lightweight and efficient intrusion detection framework capable of:

- Detecting known attacks accurately
 - Identifying unknown or zero-day attacks
 - Maintaining low computational complexity
 - Supporting lightweight IoT edge environments
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1.3 Aim of the Project

The aim of this project is to develop a lightweight hybrid intrusion detection framework capable of detecting both known and zero-day cyberattacks in IoT edge environments using Machine Learning and Deep Learning techniques.

2. MOTIVATION AND OBJECTIVE

2.1 Motivation

The rapid increase in IoT devices has introduced major security vulnerabilities due to weak security mechanisms and limited hardware resources. Traditional intrusion detection systems are generally unable to detect previously unseen attacks effectively.

During the literature survey, it was observed that many existing cybersecurity systems use complex deep learning architectures that are computationally expensive. Such models are difficult to deploy on lightweight edge devices.

This motivated the development of a lightweight hybrid intrusion detection system that combines:

- Supervised learning for known attack detection
- Anomaly detection for zero-day attack identification

while maintaining low computational overhead suitable for edge deployment.

2.2 Objectives

The major objectives of this project are:

1. To preprocess IoT network traffic data for machine learning tasks.
 2. To implement a lightweight LightGBM model for known attack classification.
 3. To implement an Autoencoder model for anomaly-based zero-day attack detection.
 4. To combine supervised and anomaly detection approaches using fusion logic.
 5. To evaluate model performance using multiple performance metrics.
 6. To design a lightweight and efficient intrusion detection framework for IoT edge devices.
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3. RELATED WORK

Several research works have explored Machine Learning and Deep Learning techniques for intrusion detection systems in IoT environments.

Traditional signature-based intrusion detection systems are effective only against previously known attack patterns and fail to identify zero-day attacks. Researchers have proposed various machine learning techniques such as Support Vector

Machine (SVM), Decision Trees, Random Forest, and XGBoost for intrusion detection.

Although these models achieve good classification performance, they generally depend on labeled attack data and struggle to identify unseen attack patterns.

Deep Learning techniques such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Autoencoders have also been widely used for anomaly detection. Autoencoders are especially useful because they learn normal traffic behavior and identify deviations as anomalies.

However, many existing deep learning-based approaches are computationally expensive and unsuitable for lightweight IoT edge deployment.

Recent research studies suggest that hybrid intrusion detection systems combining supervised classification and anomaly detection techniques can improve overall detection performance. Inspired by these approaches, this project combines LightGBM and Autoencoder models to create a lightweight hybrid intrusion detection framework.

4. PROPOSED METHODOLOGY

4.1 System Overview

The proposed system consists of four major stages:

1. Data Preprocessing
2. Known Attack Detection using LightGBM
3. Zero-Day Detection using Autoencoder
4. Fusion-Based Final Decision System

Overall workflow of the proposed system:

Input Traffic ↓ Data Preprocessing ↓ LightGBM + Autoencoder ↓ Fusion Logic ↓ Final Detection Output

4.2 Dataset Used

The Edge-IIoT dataset is used for experimentation in this project. The dataset contains normal traffic samples along with multiple attack categories commonly found in IoT environments.

Attack categories include:

- DDoS attacks
- SQL Injection
- Password attacks
- MITM attacks
- Port scanning
- Backdoor attacks
- Ransomware attacks
- XSS attacks

The dataset is suitable for intrusion detection research because it contains both normal and malicious network traffic samples.

4.3 Data Preprocessing

Data preprocessing is one of the most important stages of the project. The preprocessing pipeline was implemented using Python and Pandas.

Major preprocessing steps include:

- Removing unnecessary columns
- Removing duplicate records
- Handling missing values
- Encoding categorical features
- Normalizing feature values using StandardScaler
- Splitting known and unseen attack classes

The dataset was divided into:

- Known attack dataset
- Zero-day evaluation dataset

Known attack classes were used for LightGBM training, while selected unseen attack classes were reserved for zero-day evaluation.

4.4 LightGBM Model

LightGBM is used as the supervised classification model for known attack detection.

Reasons for Selecting LightGBM

- Lightweight architecture
- Fast training speed
- Low memory consumption
- High classification performance
- Suitable for edge deployment

The model was trained using labeled known attack classes and normal traffic data.

Working Principle

LightGBM learns patterns associated with known attack categories and performs multiclass classification during test